ISMAIL ALTAY ATAMAN

github.com/AltayAtaman linkedin.com/in/ismailaltayataman i.altay.ataman@gmail.com

SUMMARY

Software Engineer with 3 years of professional experience in IoT, embedded, and full-stack systems. Specializes in C/C++, IoT architecture, AWS IoT Core, Azure IoT Hub, and Flutter. Experienced in multi-node CAN bus protocols, fail-safe OTA pipelines, diagnostic tools, and production-ready applications. Builds reliable, field-ready products using AI-assisted development workflows.

WORK EXPERIENCE

Embedded and IoT Solutions Engineer, Nerva / Lemi Group Inc. — October 2023 – August 2026

- Architected scalable telemetry systems using AWS IoT Core and Azure IoT Hub with automated Fleet Provisioning, supporting real-time monitoring across a growing device fleet.
- Developed C++ firmware for cellular IoT devices, implementing fail-safe OTA updates that significantly reduced on-site field maintenance interventions.
- Designed a custom multi-node CAN bus protocol spanning a three-zone hardware architecture for a mobility device, coordinating real-time communication between motor controllers, an IoT gateway, and an SBC-based HMI.
- Built cross-platform HMI and diagnostic tools using Flutter targeting embedded Linux (Rockchip RK3562), visualizing low-level hardware telemetry for a wheelchair control system.

Intern Engineer, Baykar Technologies — July– August 2023

- Developed automated test scripts using Winium to verify UAV ground-station mission-control software, significantly reducing manual regression testing time.

Intern Engineer, DeltaV Space Technologies — January – February 2023

- Designed and tuned discrete PID control algorithms for a 3-axis reaction wheel assembly, stabilizing payload orientation under simulated disturbances.

Intern Engineer, TÜBİTAK SAGE — August - September 2022

- Built GUIs for embedded defense systems using Qt and C++, improving operator interaction efficiency.

PROJECTS

- **yiy.io | Offline-First POS & Automation Platform (Project Page)**
  - Architected a zero-cloud-dependency, offline-first POS platform (Node.js/SQLite) running locally to ensure uninterrupted venue operations and zero cloud latency.
  - Built a React/TypeScript admin dashboard for real-time order analytics alongside a Flutter waiter app that pairs via local Wi-Fi QR sync.
  - Packaged full-stack architecture into a single standalone executable (.exe) for one-click deployment without external database setup.
- **Yildiz Rocket Team | Team Leader & Avionics Lead**
  - Achieved Finalist status at Spaceport America Cup, leading a 50+ person cross-functional team to design and launch high-altitude rockets reaching 20,000 ft.
  - Designed the avionics architecture and control software for hybrid rocket motor propulsion, integrating real-time sensor fusion and parachute deployment logic.
- **TÜBİTAK 2209A Project:** Modeled control systems for spacecraft atmospheric landing simulations using MATLAB/Simulink.

SKILLS

- **Languages:** Turkish (Native), English (C1)
- **Programming:** C, C++, C#, Flutter/Dart, Python, JavaScript/TypeScript (Node.js/Automation), MATLAB
- **Embedded Systems & Hardware:** ESP32 (ESP-IDF), STM32, FreeRTOS, Cellular Modems (GSM/LTE/Quectel), GNSS
- **Protocols & Architecture:** CAN Bus 2.0B, UART, SPI, I2C, TCP/IP, MQTT, WebSockets, RESTful APIs, Local Network Sync
- **Cloud & IoT:** AWS IoT Core, Azure IoT Hub, Fleet Provisioning, OTA Update Architecture, SQLite (Edge Storage)
- **Tools & DevOps**: Git, GitHub Actions (CI/CD), CMake, Linux CLI, Altium Designer, VS Code
- **Domain Expertise:** Embedded/IoT Systems, Mobility/Control Systems, Avionics, Full-Stack Local Automation

EDUCATION

Yildiz Technical University (ISTANBUL/TURKEY)

- Mechatronics Engineering - 2025